Assignment 8

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1 Experimental Setup

This assignment reuses the salt image segmentation task from the previous ResNet/Bayesian assignment. The dataset contains paired seismic-like images and binary masks, where the goal is to predict salt regions at the pixel level. In this Week 10 experiment, the same segmentation task is used to compare a residual CNN encoder–decoder model with a Vision Transformer based segmentation model under similar training conditions.

1.1 Dataset and Shared Training Conditions

Item	Setting
Data source	`ResNet_and_Bayesian/Salt`
Task type	Binary image segmentation
Input image size	64×64
Total number of image-mask pairs	4000
Train / validation / test split	2890 / 510 / 600
Positive pixel fraction in the training masks	0.2455
Number of epochs	60
Batch size	32
Loss function	`BCEWithLogitsLoss`
Evaluation metrics	Pixel Accuracy, F1-score/Dice, RMSE, and Mean IoU

Table 1: Dataset configuration and shared training conditions used for both models.

To ensure a fair comparison, both models use the same dataset split, input resolution, batch size, number of training epochs, loss function, and evaluation metrics.

1.2 Model Configurations

Model	Architecture and Key Settings	Parameters	Optimizer	Learning Rate / Weight Decay
ResidualSegNet CNN	Residual convolutional encoder–decoder adapted from the previous segmentation assignment	516,545	Adam	1×10^{-3} / 1×10^{-5}
Vision Transformer Segmenter	ViT encoder with a convolutional upsampling decoder; patch size 4×4, transformer dim 96, depth 4, heads 4, MLP dim 192, decoder channels 96/48	588,673	AdamW	3×10^{-4} / 1×10^{-4}

In terms of model size, the ViT has slightly more parameters than the CNN, with 72,128 additional parameters, which is approximately a 14% increase.

2 Results

2.1 Validation Results

Model	Best Validation Mean IoU	Training Time (s)
ResidualSegNet CNN	0.6974	724.0
Vision Transformer	0.6063	1254.6

The validation results show that ResidualSegNet CNN outperforms the Vision Transformer by 0.0911 in best mean IoU, while also requiring substantially less training time. Compared with the CNN, the ViT takes 530.6 more seconds to train, which is about 73% slower.

2.2 Test Results

Model	Pixel Accuracy	F1-score / Dice	RMSE	Mean IoU
ResidualSegNet CNN	0.9225	0.7468	0.2422	0.7186
Vision Transformer	0.8647	0.6580	0.3211	0.6159

On the test set, ResidualSegNet CNN performs better than the Vision Transformer across all four metrics:

- Pixel Accuracy improves by 0.0578;
- F1-score / Dice improves by 0.0888;
- Mean IoU improves by 0.1027;
- RMSE is reduced by 0.0789.

These results indicate that, under the current experimental setup, the CNN is not only more accurate overall, but also achieves better overlap-based segmentation performance on the test set, as reflected by its higher Dice and mean IoU scores and lower RMSE.

2.3 Result Visualizations

To complement the quantitative comparison above, Figure 1 presents the training-history comparison between the two models, while Figure 2 provides qualitative segmentation examples from representative samples.

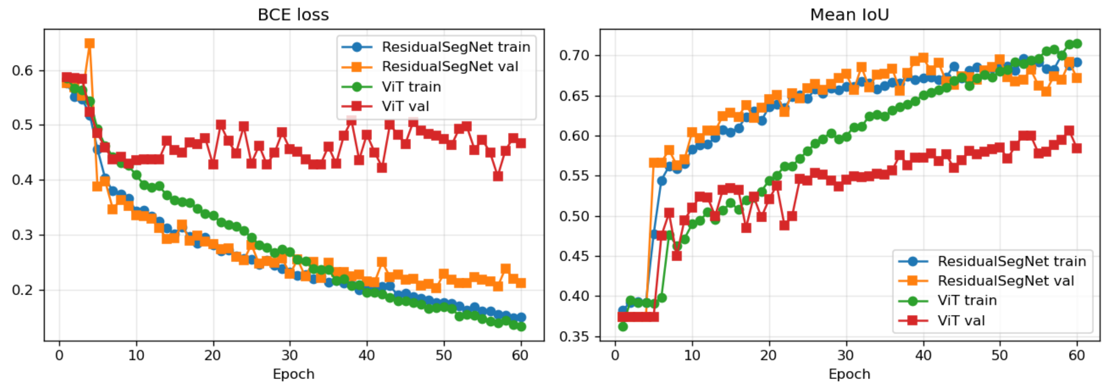

Figure 1: Training-history comparison for ResidualSegNet CNN and the Vision Transformer. The left panel shows the loss curves across epochs, while the right panel shows the segmentation-performance curves over the same training process.

3 Analysis and Discussion

3.1 Improvements and Limitations of ViT

The experiment notes that reducing the ViT patch size from 8×8 to 4×4 improves performance. This is a reasonable outcome because smaller patches preserve more spatial detail, and segmentation tasks rely heavily on boundaries, textures, and local structural information. As a result, finer-grained tokens are generally more helpful for accurate mask prediction.

However, even after this improvement, the ViT still underperforms compared with ResidualSegNet CNN. Several factors may explain this result:

- Salt segmentation depends strongly on local texture and edge information, and convolutional models naturally provide a strong locality bias;
- The residual encoder–decoder structure is effective at both extracting features and reconstructing spatial structure;
- A ViT trained from scratch often requires more data, stronger regularization, or pretraining in order to reach competitive performance;
- The current dataset size of 4000 samples is still relatively small for a transformer-based model.

3.2 Training Stability

The experiment also shows that the ViT training curve continues to improve on the training set, while validation performance becomes unstable in later epochs. This suggests that simply training longer may not reliably close the gap with the CNN and may instead worsen generalization stability. More promising directions for improvement include:

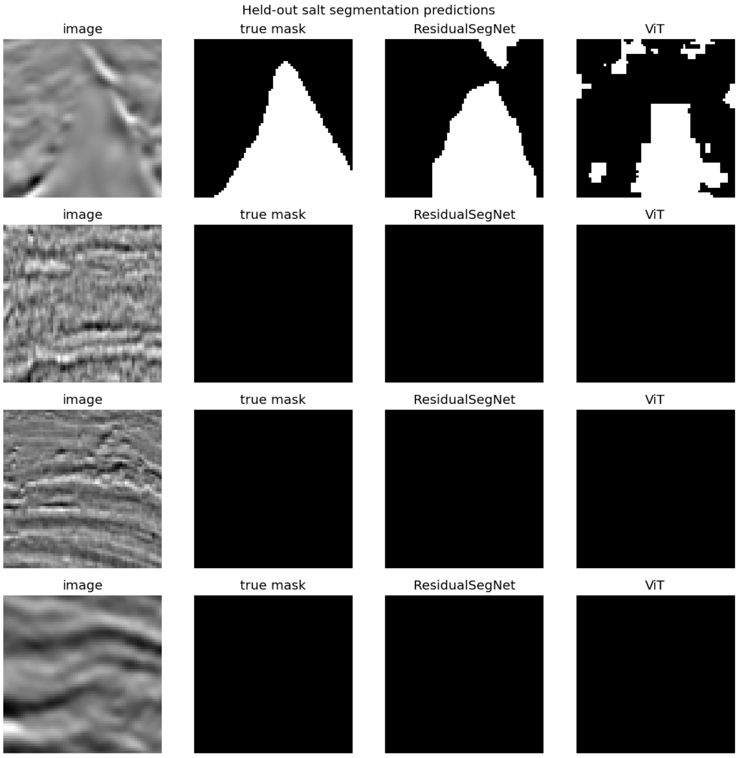

Figure 2: Qualitative comparison on representative samples. Each row shows one example, and the columns correspond to the input image, the ground-truth mask, the ResidualSegNet CNN prediction, and the Vision Transformer prediction.

- using a pretrained transformer backbone;
- applying stronger data augmentation;
- adopting a transformer architecture designed specifically for segmentation;
- adding explicit skip connections between encoder and decoder stages.

4 Conclusion

Based on the validation and test results, the following conclusions can be drawn:

1. Under the current salt segmentation setup, **ResidualSegNet CNN is the stronger model**;
2. The Vision Transformer benefits from the smaller 4×4 patch size, which highlights the importance of spatial resolution for segmentation;
3. Even with this improvement, the ViT still performs worse than the CNN on all reported test metrics, while also requiring more training time;
4. Future improvements to the ViT should focus on pretraining, architectural refinement, and stronger augmentation rather than simply extending the number of epochs.

Overall, this experiment shows that for a small-scale salt segmentation task, a convolutional network with a strong locality bias remains more effective. Although the Vision Transformer demonstrates potential, it is still difficult for it to surpass a well-designed CNN baseline without more suitable training strategies or model adaptations.

5 Code

https://github.com/WKKKKKKKKKKKKKK/ML_IN_GEO